

Physics of Semiconductor Devices

ECE210: PSD



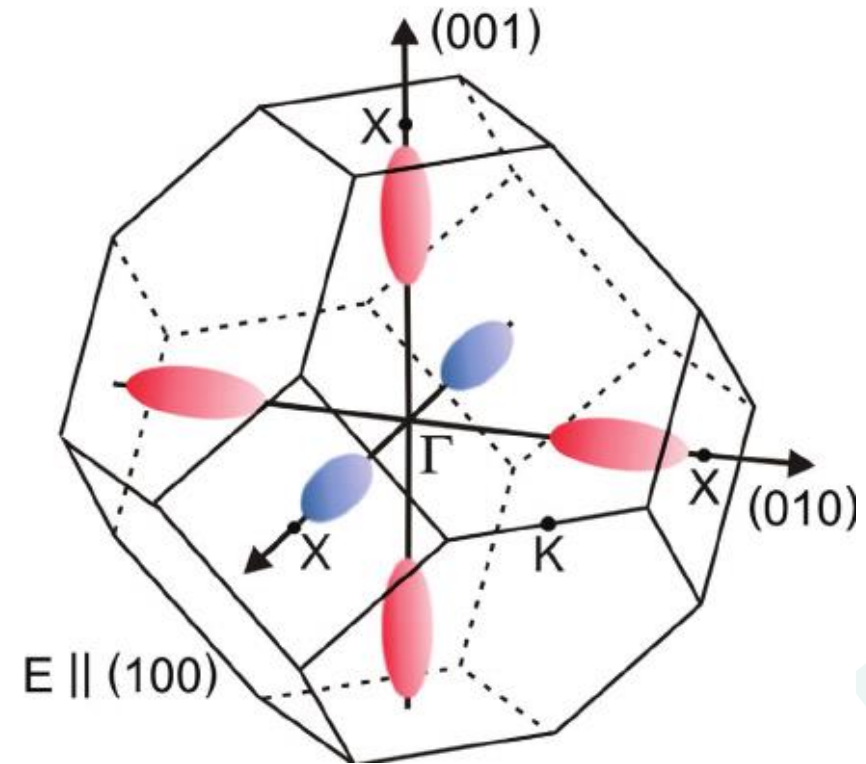
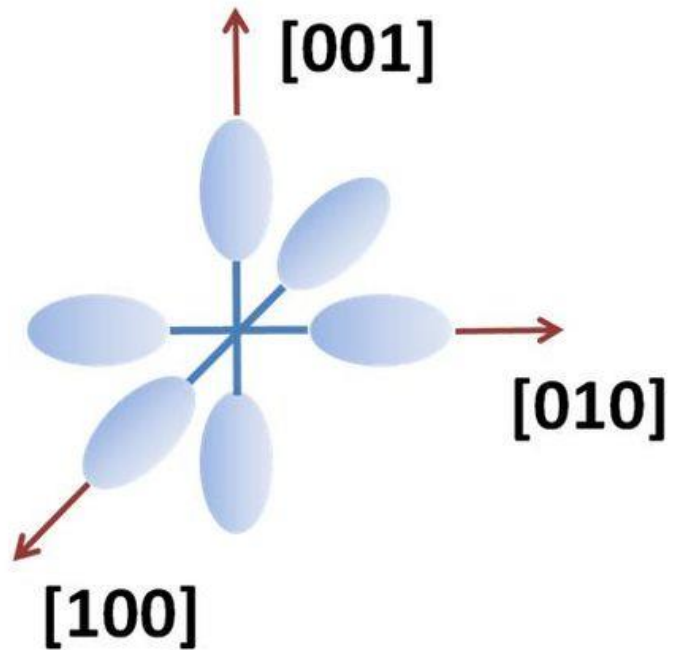
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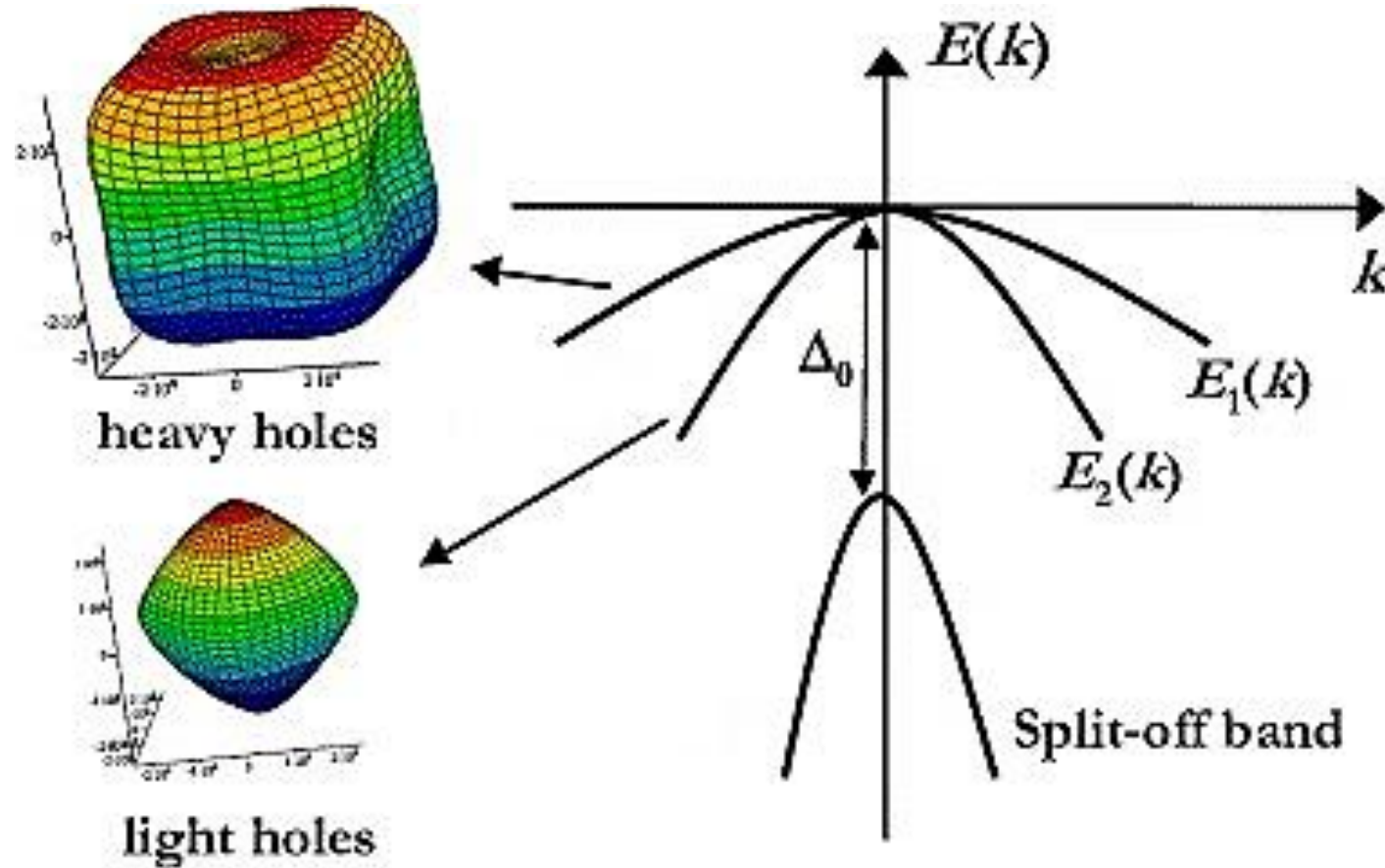
Week 07

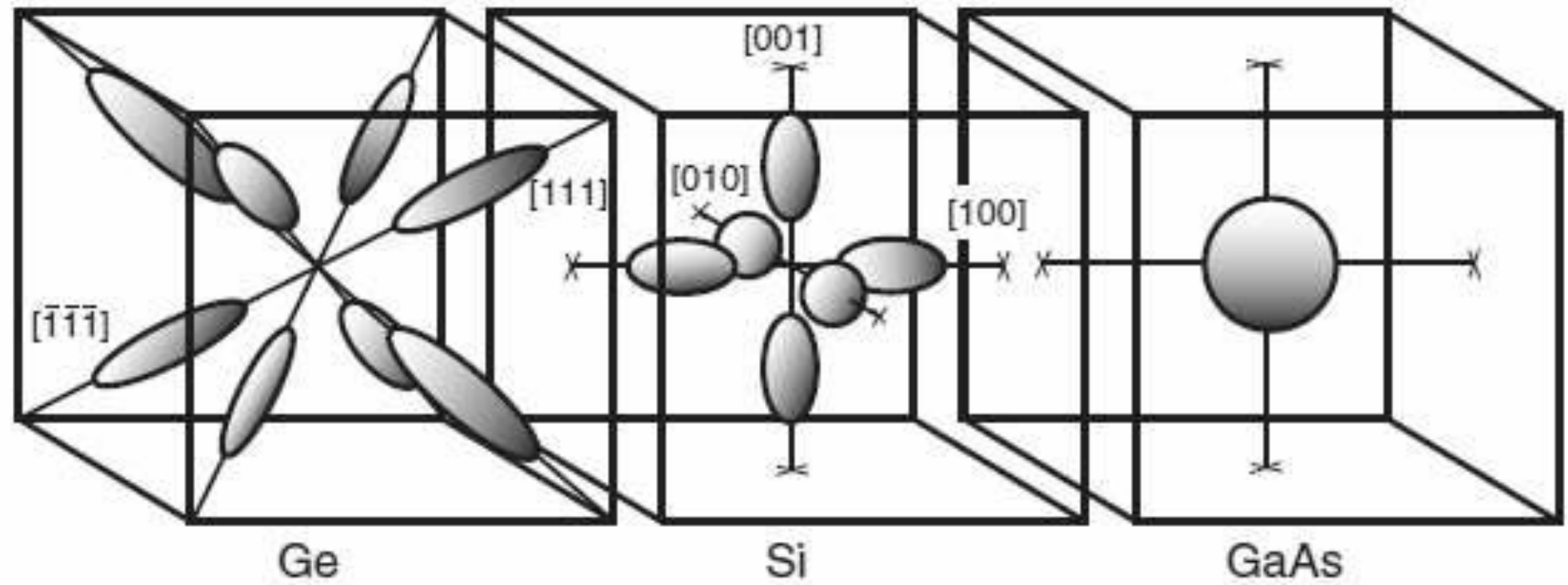


Constant electron energy surface



Constant hole energy surface of Si





Parabolic approximation in different valleys

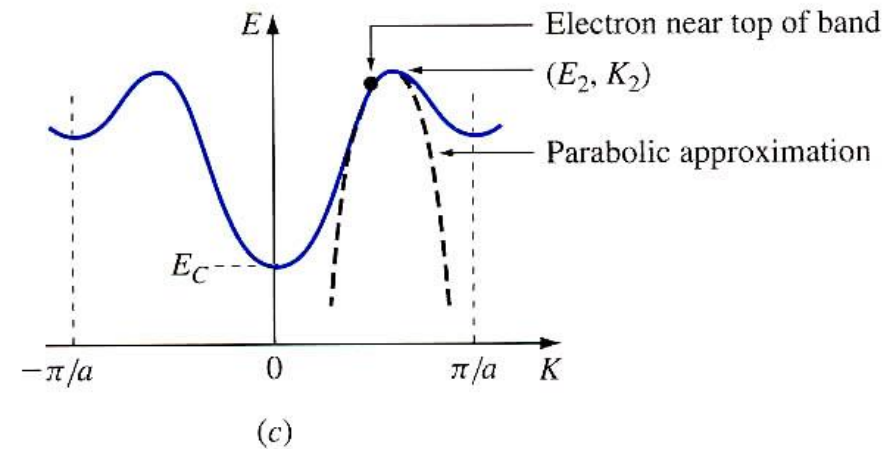
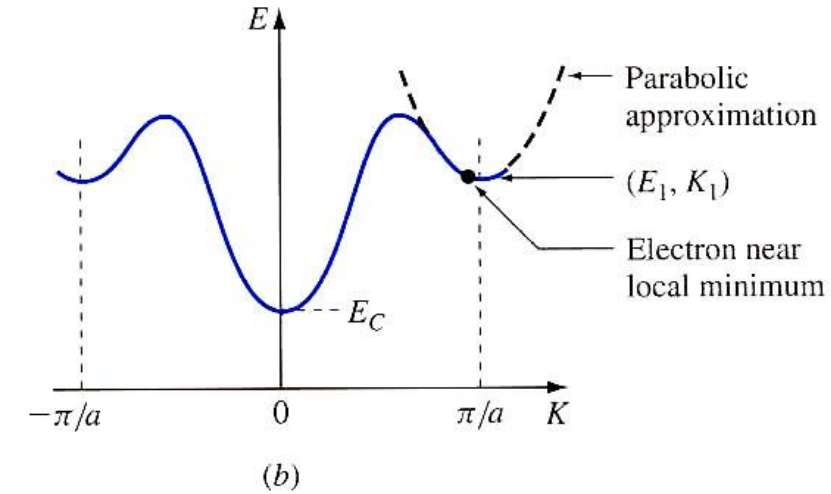
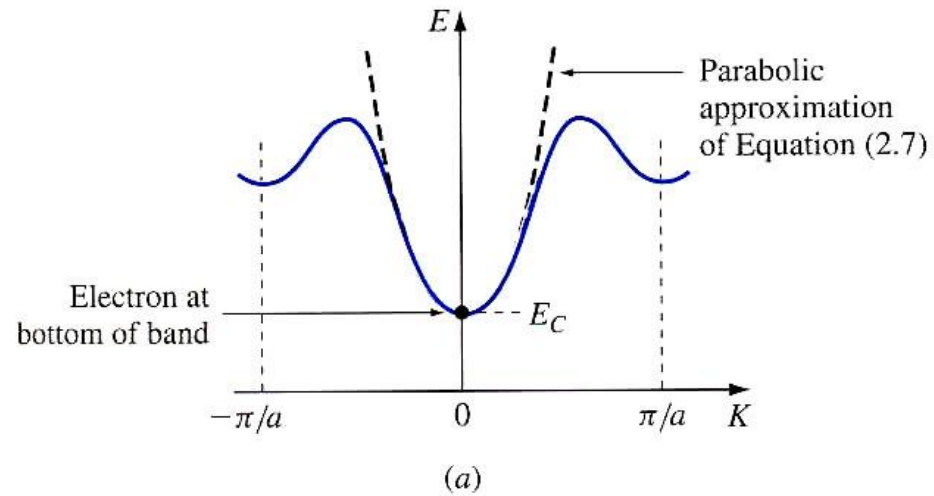
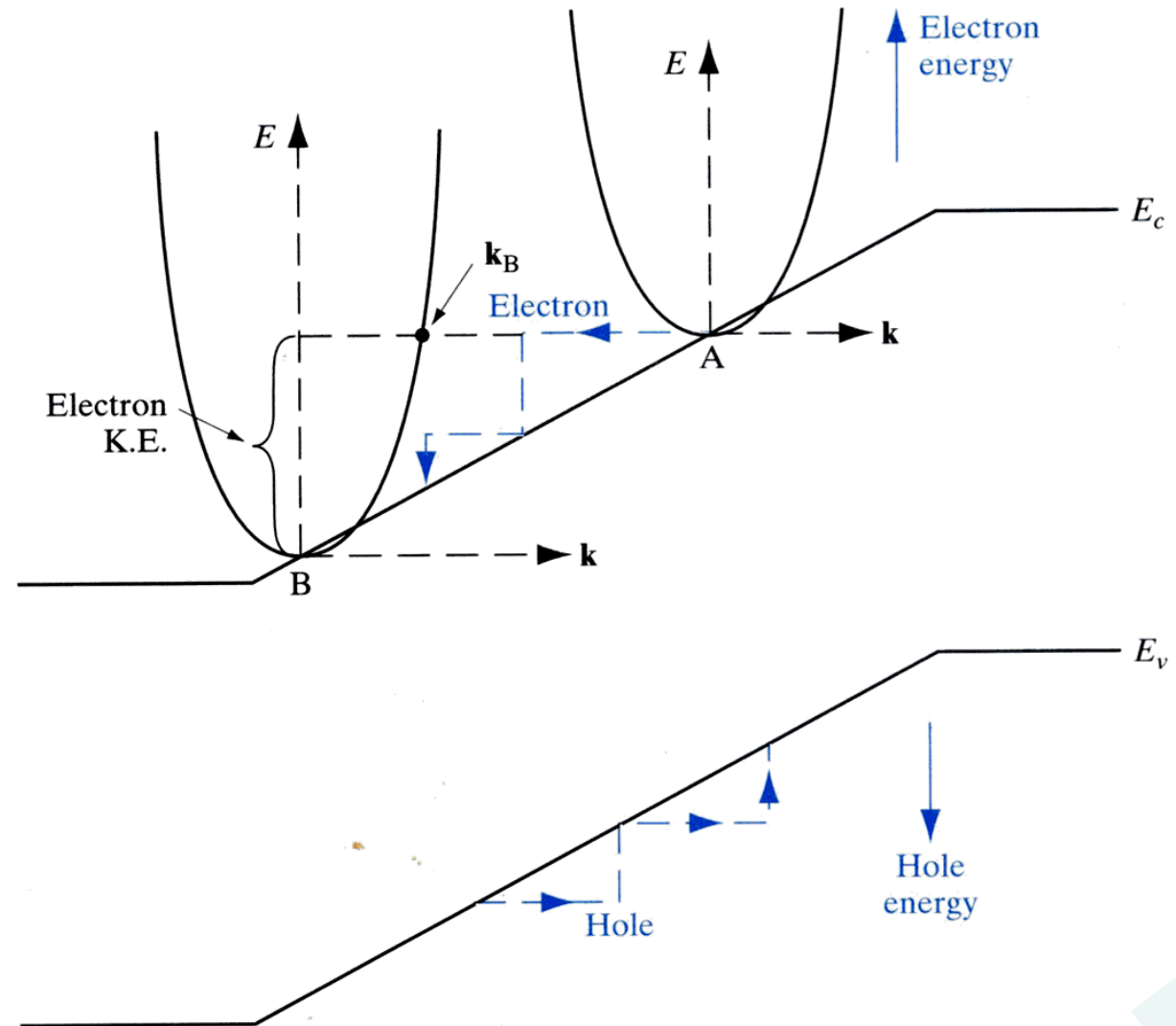
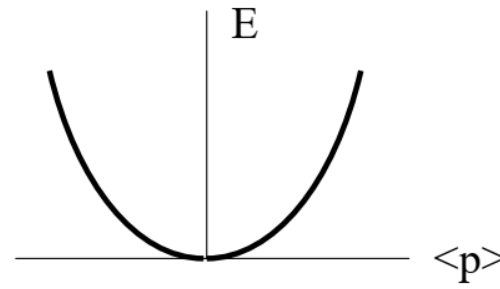


Figure 3-9

Superimposition of the $(E, \mathbf{k})$ bandstructure on the E -versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wavevectors point in opposite directions and these charge carriers move opposite to each other, as shown.



$$(1) \quad E = \frac{\hbar^2 k^2}{2m_0} = \frac{p^2}{2m_0} \quad (2)$$



(3)

$$v_g = \frac{dE}{dp} = \frac{1}{\hbar} \frac{dE}{dk}$$

Electron Dynamics in the Lattice

The work done by the force on the wavepacket will then be

$$(4) \quad dE = Fdx = Fv_g dt$$

From that we get the force expression using (3)

$$(5) \quad F = \frac{1}{v_g} \frac{dE}{dt} = \frac{1}{v_g} \frac{dE}{dk} \frac{dk}{dt}$$

$$(6) \quad F = \frac{d(\hbar k)}{dt}$$

The acceleration is found taking time derivative of (3)

$$(7) \quad a = \frac{dv_g}{dt} = \frac{1}{\hbar} \frac{d}{dt} \left(\frac{dE}{dk} \right) = \frac{1}{\hbar^2} \left(\frac{d^2 E}{dk^2} \right) \frac{d(\hbar k)}{dt}$$

Effective Mass Expression

Finally, we obtain the effective mass equation

$$(8) \quad F = m^* \cdot \frac{dv_g}{dt}$$

$$(9) \quad m^* = \frac{1}{\frac{1}{\hbar^2} \frac{d^2 E}{dk^2}}$$

The equation (8) is identical to Newton's second law of motion except that the actual particle mass is replaced by an *effective mass* m^* .

Effective mass



Curvature of the band determines the effective mass of the carriers in a crystal, which is different from the free electron mass.

- For **parabolic bands**, the components of the effective mass tensor are calculated according to:

$$\frac{1}{m_{ij}^*} = \frac{1}{\hbar^2} \cdot \frac{\partial^2 E}{\partial k_i \partial k_j} \quad \xrightarrow{\text{Si}} \quad \frac{1}{m^*} = \begin{bmatrix} \frac{1}{m_{xx}^*} & 0 & 0 \\ 0 & \frac{1}{m_{yy}^*} & 0 \\ 0 & 0 & \frac{1}{m_{zz}^*} \end{bmatrix}$$

Effective mass of Si/Ge/GaAs



Name	Symbol	Germanium	Silicon	Gallium Arsenide
Band minimum at $k = 0$				
Minimum energy	$E_{g,direct}$ [eV]	0.8	3.2	1.424
Effective mass	m_e^*/m_0	0.041	~0.2?	0.067
Band minimum <i>not</i> at $k = 0$				
Minimum energy	$E_{g,indirect}$ [eV]	0.66	1.12	1.734
Longitudinal effective mass	$m_{e,l}^*/m_0$	1.64	0.98	1.98
Transverse effective mass	$m_{e,t}^*/m_0$	0.082	0.19	0.37
Wavenumber at minimum	k [1/nm]	xxx	xxx	xxx
Longitudinal direction		(111)	(100)	(111)
Heavy hole valence band maximum at $E = k = 0$				
Effective mass	m_{hh}^*/m_0	0.28	0.49	0.45
Light hole valence band maximum at $k = 0$				
Effective mass	m_{lh}^*/m_0	0.044	0.16	0.082
Split-off hole valence band maximum at $k = 0$				
Split-off band valence band energy	$E_{v,so}$ [eV]	-0.028	-0.044	-0.34
Effective mass	$m_{h,so}^*/m_0$	0.084	0.29	0.154

Carrier densities and density of states



As we eventually want to describe the current-voltage characteristics of a device, the next important step is to determine the carriers (electron and hole) that are available for the conduction.

The number of carriers is a function of available energy states or the number of quantum states. (as per the Pauli exclusion principle, the two electrons (up + down spin) can occupy one energy state).

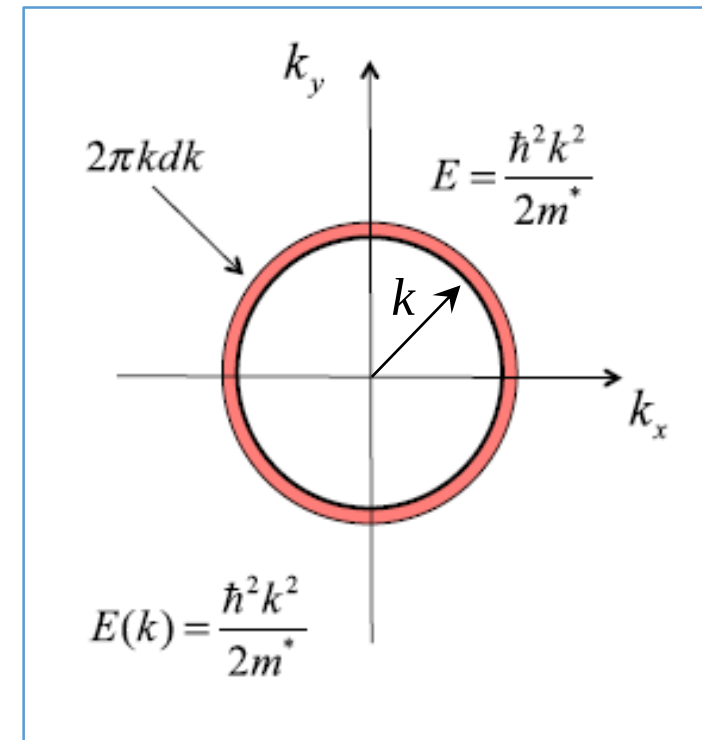
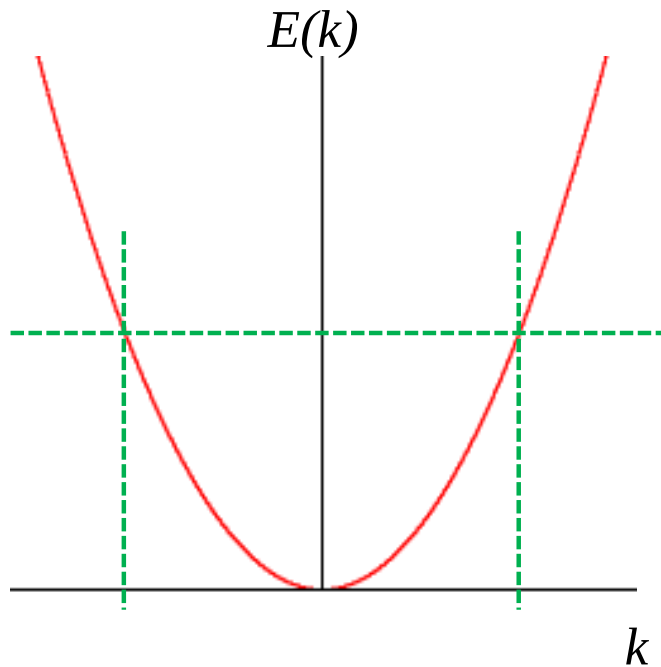
The number of quantum states per energy (as we can consider an energy band is actually made of closely packed discrete energy levels) per unit dimension is nothing but the **Density of States**.

DOS in 2D (using parabolic approximation)

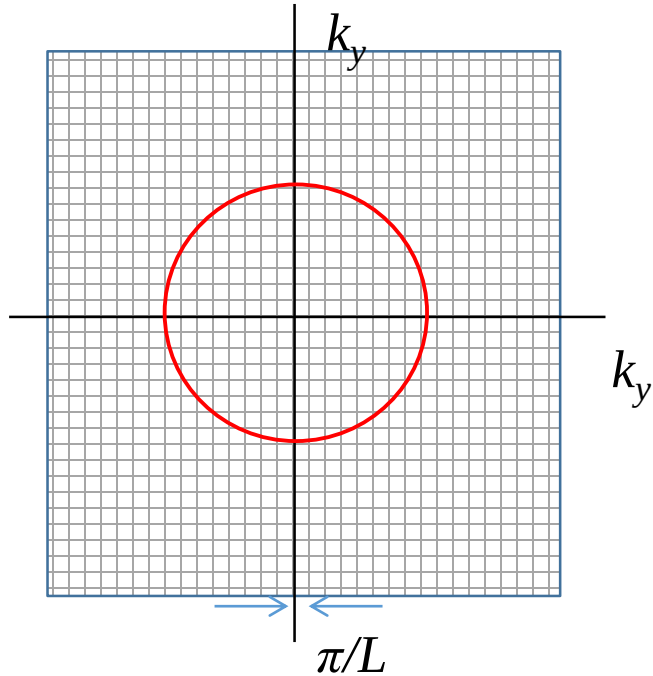
DOS (E) = number of states **per energy** per unit volume (for 3D)

DOS (E) = number of states **per energy** per unit area (for 2D)

DOS (E) = number of states **per energy** per unit line (for 1D)



DOS in 2D (using parabolic approximation)



number of electrons in a 2-dimensional circle is given by

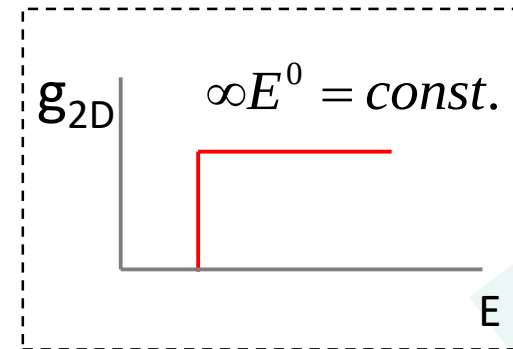
$$N_{2D} = \frac{\pi k^2}{4 \left(\frac{\pi}{L} \right)^2} \times 2 = \frac{k^2 L^2}{2\pi} \Rightarrow N_{2D} = \frac{m^* E}{\pi \hbar^2} L^2$$

where $k^2 = k_x^2 + k_y^2$ and $E = \frac{\hbar^2 k^2}{2m^*}$

Therefore, the density of states will be (no. of electron/energy/area)

$$g_{2D}(E) = \frac{\partial N_{2D}}{\partial E} \times \frac{1}{L^2}$$

$$g_{2D}(E) = \frac{m^*}{\pi \hbar^2}$$



Carrier density $n = \int_0^{\infty} g(E) f(E) dE$

DOS in 3D, 2D, 1D, 0D



no confinement

3D materials
(bulk)

$$g_{3D}(E) = g_v \frac{(2m^*)^{3/2}}{2\pi^2 \hbar^3} \sqrt{E - E_c}$$

1D confinement

2D materials
Quantum well/thin
film

$$g_{2D}(E) = g_v \frac{m^*}{\pi \hbar^2}$$

2D confinement

1D materials
quantum
wire/nanotube

$$g_{1D}(E) = g_v \frac{1}{\pi \hbar} \sqrt{\frac{m^*}{2(E - E_c)}}$$

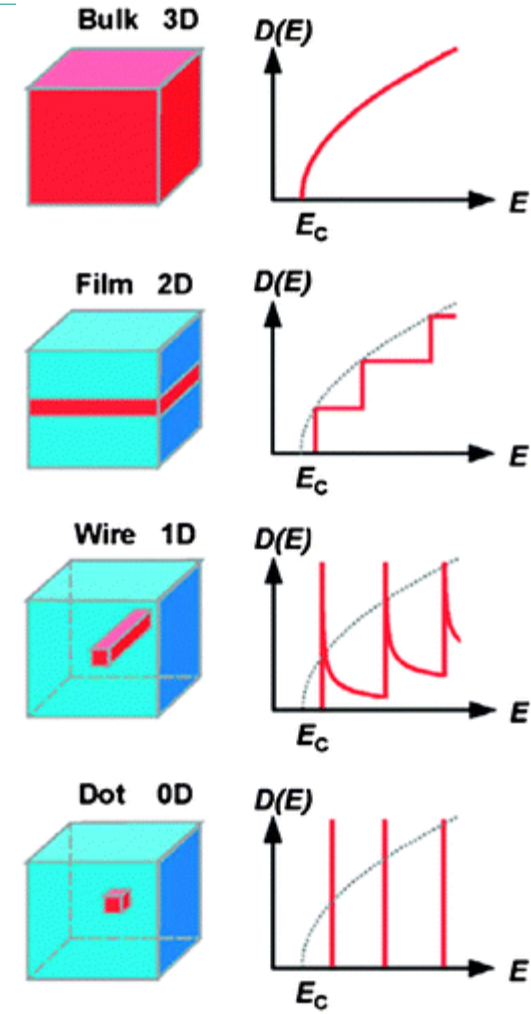
3D confinement

0D materials
quantum dot
/nanoparticle

$$g_{0D}(E) = g_v 2\delta(E - E_c)$$

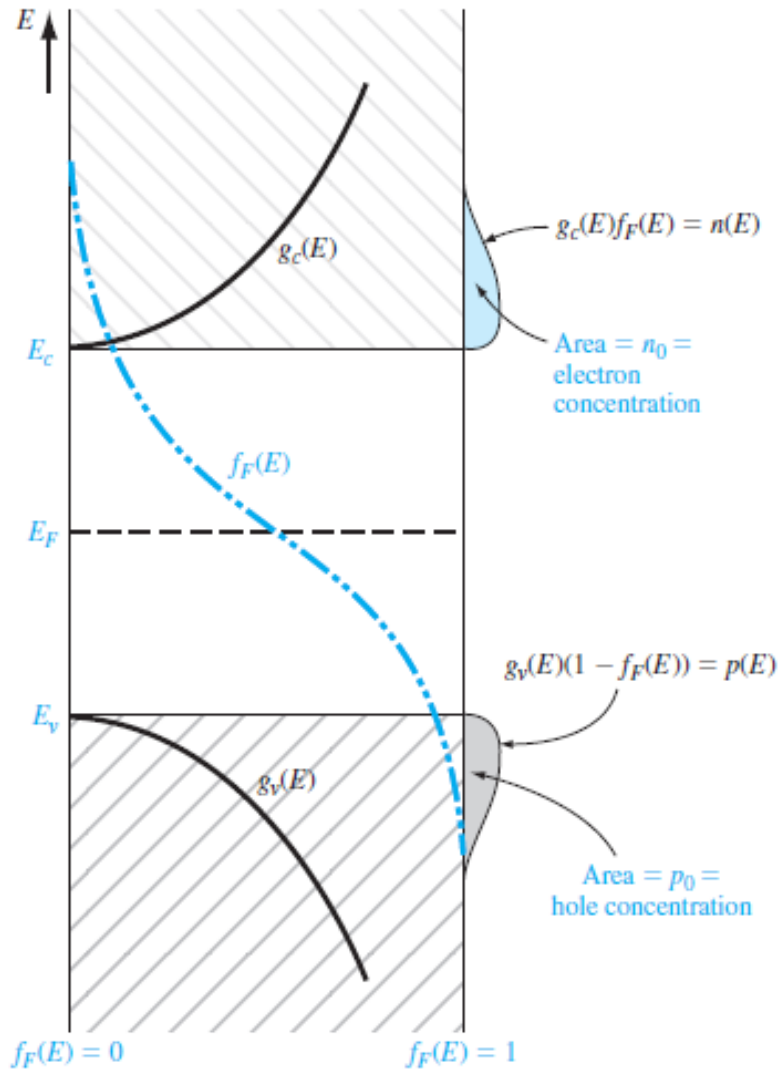
where g_v is the valley degeneracy

$E > E_c$

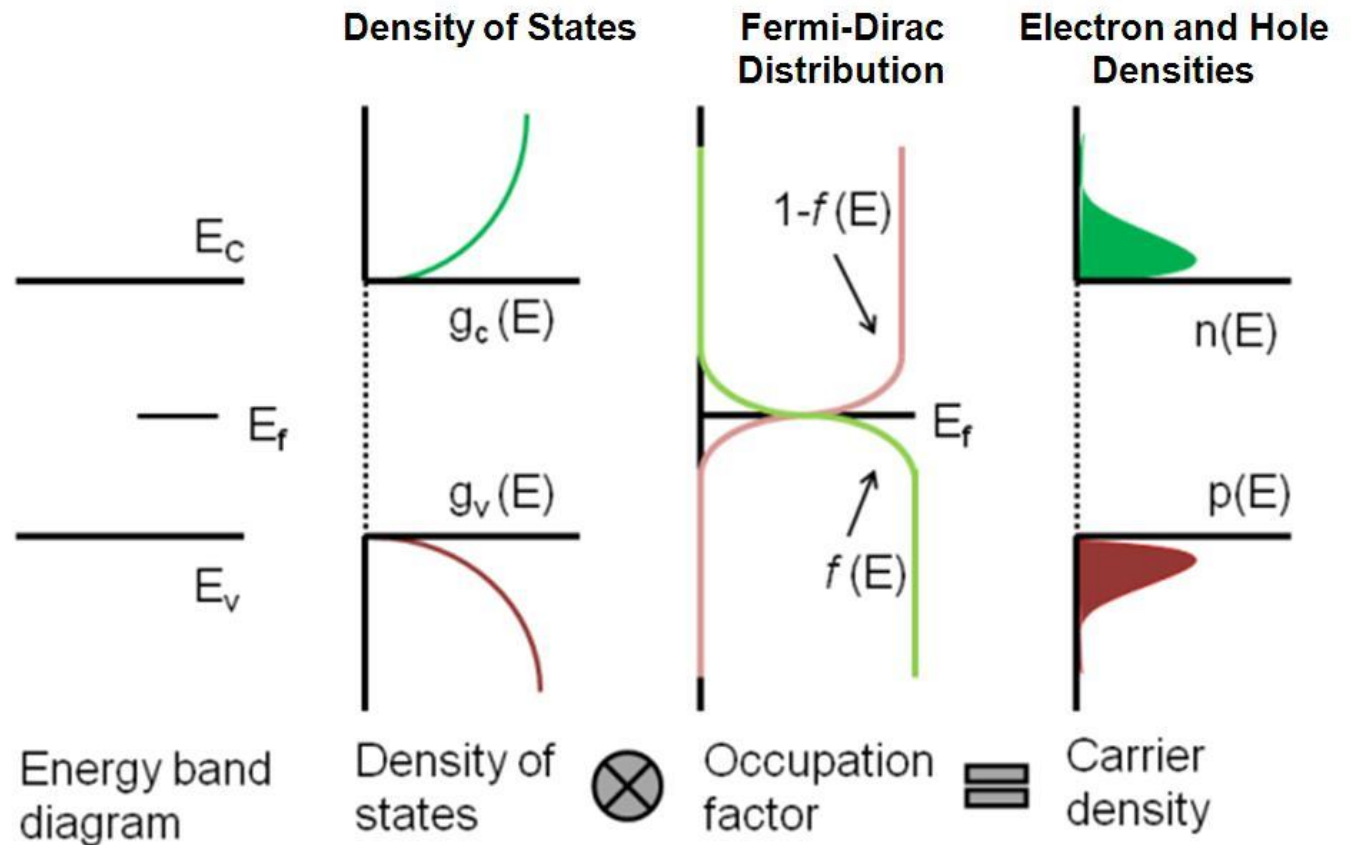


Try yourself to derive the expression of DOS in 3D, 2D, 1D, 0D with the non-parabolic band

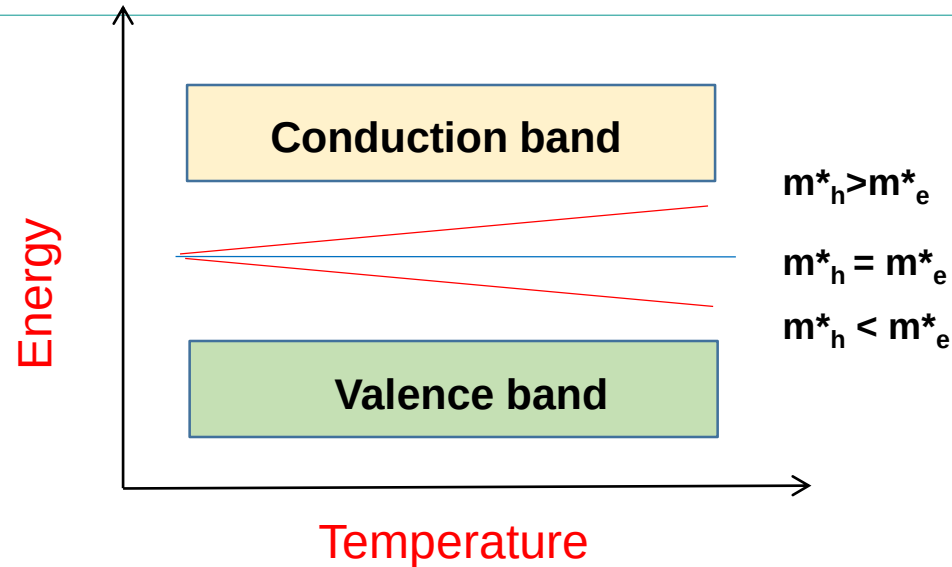
Intrinsic semiconductor and carrier concentration



Plots of $D(E)$, $f(E)$, n , and p as a Function of Energy



Fermi level in Intrinsic semiconductor



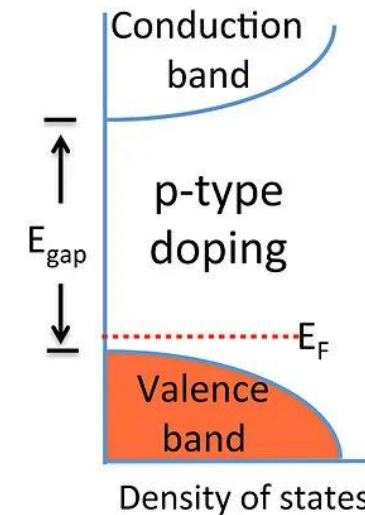
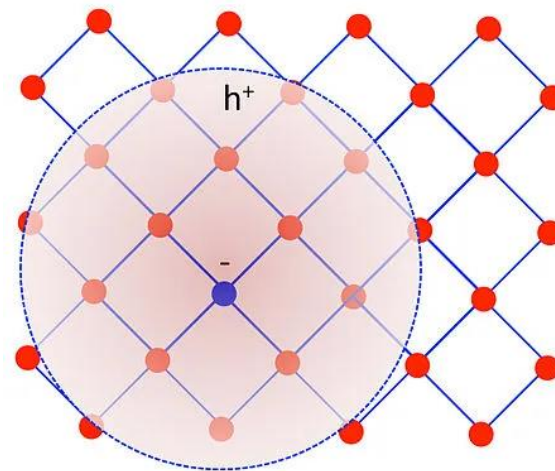
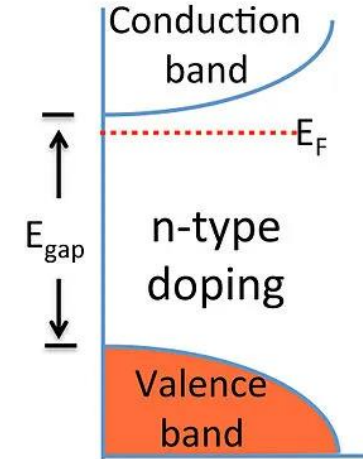
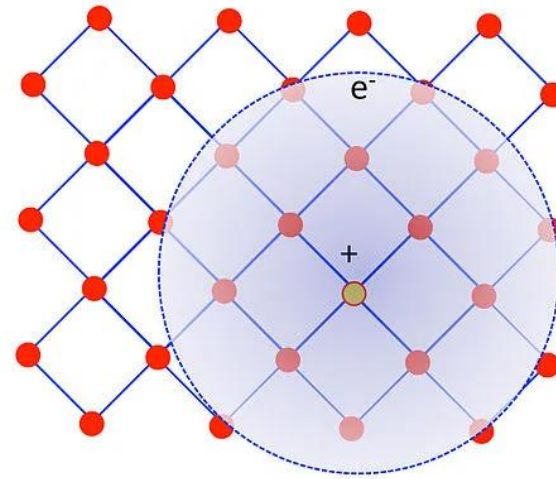
- At $T=0K$ the number of electrons in conduction band equals the number of holes in valence band, $n_e = n_h = n_i$.

Fermi level lies in the middle of the band gap.

- At $T \neq 0 K$, it depends upon the effective mass of electron and hole.

$$E_f = \left(\frac{E_g}{2} \right) + \frac{3}{4} kT \ln \frac{m_h^*}{m_e^*}$$

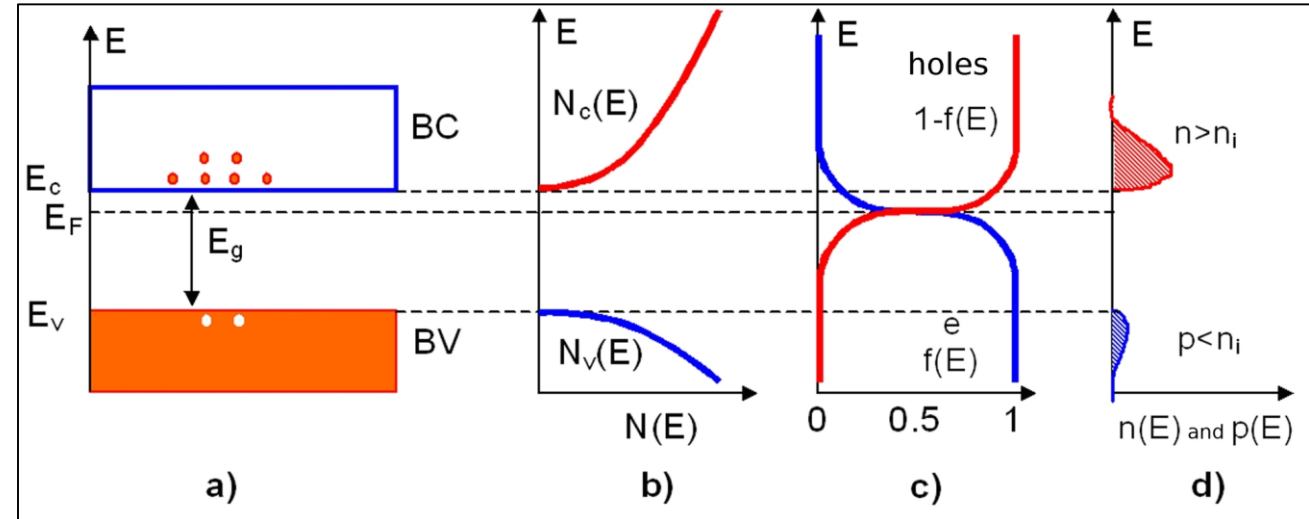
Extrinsic semiconductor (doped)



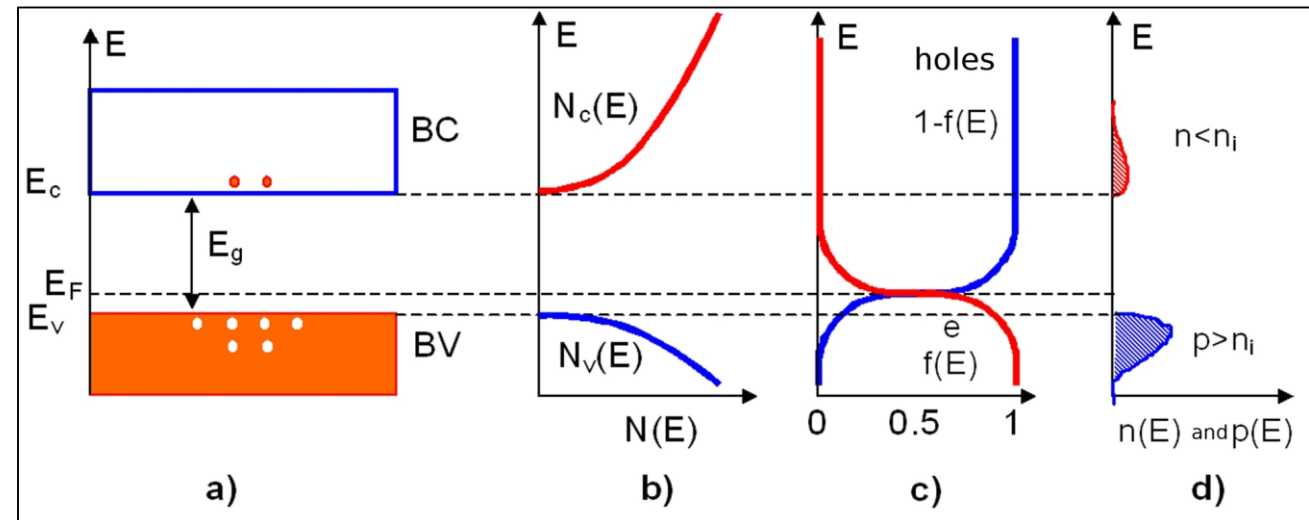
Extrinsic semiconductor (doped) and carrier concentration



n-type semiconductor



p-type semiconductor

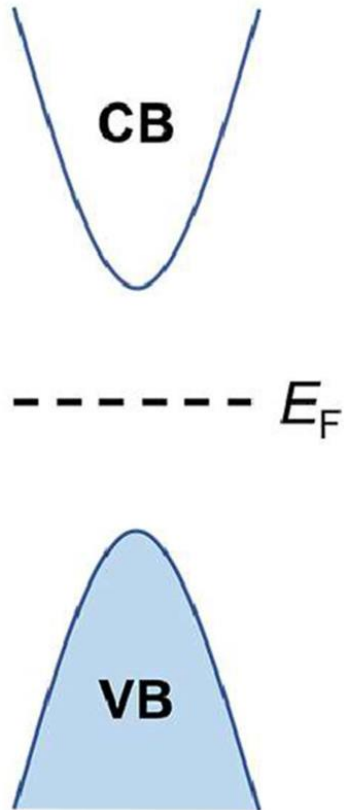


We will derive the expression of carrier densities after the mid sem

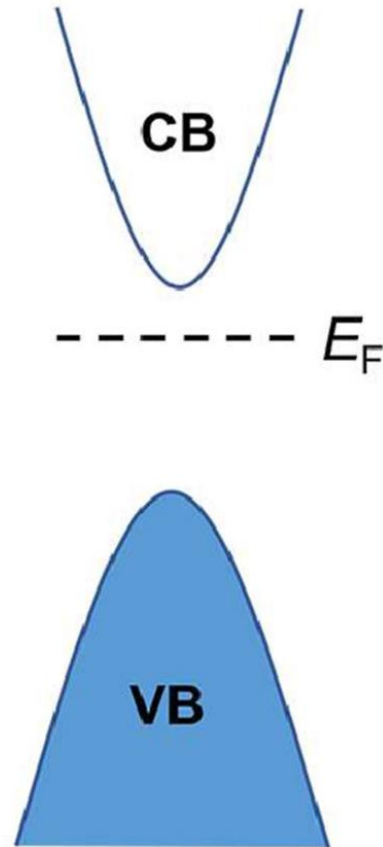
Non-degenerate and degenerate semiconductor



Intrinsic



Extrinsic non-degenerate n-type



Extrinsic degenerate n-type

